

Differential inequalities and Taylor estimates

Calculus and analysis — Session 2

Vahid Shahverdi

TU/e IMC team

Integrating factors

Let $u \in C^1([a, b])$, and let p, q be continuous. Suppose

$$u'(x) - p(x)u(x) \geq q(x).$$

Put

$$A(x) = \int_a^x p(t) \, dt.$$

Integrating factors

Let $u \in C^1([a, b])$, and let p, q be continuous. Suppose

$$u'(x) - p(x)u(x) \geq q(x).$$

Put

$$A(x) = \int_a^x p(t) \, dt.$$

Then

$$(e^{-A(x)}u(x))' \geq e^{-A(x)}q(x).$$

Integrating factors

Let $u \in C^1([a, b])$, and let p, q be continuous. Suppose

$$u'(x) - p(x)u(x) \geq q(x).$$

Put

$$A(x) = \int_a^x p(t) \, dt.$$

Then

$$(e^{-A(x)}u(x))' \geq e^{-A(x)}q(x).$$

Integrating from a to $x \geq a$,

$$u(x) \geq e^{A(x)} \left(u(a) + \int_a^x e^{-A(t)} q(t) \, dt \right).$$

Gronwall's inequality: a basic form

Gronwall's inequality

Let $F \in C^1([a, b])$ satisfy

$$F'(x) \leq \lambda F(x).$$

Then, for $a \leq x \leq b$,

$$F(x) \leq F(a)e^{\lambda(x-a)}.$$

Gronwall's inequality: a basic form

Gronwall's inequality

Let $F \in C^1([a, b])$ satisfy

$$F'(x) \leq \lambda F(x).$$

Then, for $a \leq x \leq b$,

$$F(x) \leq F(a)e^{\lambda(x-a)}.$$

Proof.

$$(e^{-\lambda(x-a)}F(x))' = e^{-\lambda(x-a)}(F'(x) - \lambda F(x)) \leq 0.$$

Gronwall's inequality: a basic form

Gronwall's inequality

Let $F \in C^1([a, b])$ satisfy

$$F'(x) \leq \lambda F(x).$$

Then, for $a \leq x \leq b$,

$$F(x) \leq F(a)e^{\lambda(x-a)}.$$

Proof.

$$(e^{-\lambda(x-a)} F(x))' = e^{-\lambda(x-a)} (F'(x) - \lambda F(x)) \leq 0.$$

In particular,

$$F \geq 0, \quad F(a) = 0, \quad F' \leq \lambda F \quad \implies \quad F \equiv 0.$$

Example 1: IMC 1994, Problem 7

Let $f \in C^1([a, b])$, with $f(a) = 0$. Suppose that, for some $\lambda > 0$,

$$|f'(x)| \leq \lambda |f(x)| \quad (a \leq x \leq b).$$

Must f vanish throughout $[a, b]$?

Example 1: IMC 1994, Problem 7

Let $f \in C^1([a, b])$, with $f(a) = 0$. Suppose that, for some $\lambda > 0$,

$$|f'(x)| \leq \lambda |f(x)| \quad (a \leq x \leq b).$$

Must f vanish throughout $[a, b]$?

Auxiliary function:

$$F(x) = \int_a^x |f(t)| \, dt.$$

Notes: I1994-2-1. IMC 1994, Day 2, Problem 1.

Example 1: solution

Since $f(a) = 0$,

$$|f(x)| = \left| \int_a^x f'(t) \, dt \right| \leq \lambda \int_a^x |f(t)| \, dt = \lambda F(x).$$

Example 1: solution

Since $f(a) = 0$,

$$|f(x)| = \left| \int_a^x f'(t) \, dt \right| \leq \lambda \int_a^x |f(t)| \, dt = \lambda F(x).$$

Therefore

$$F'(x) = |f(x)| \leq \lambda F(x), \quad F(a) = 0, \quad F \geq 0.$$

Example 1: solution

Since $f(a) = 0$,

$$|f(x)| = \left| \int_a^x f'(t) \, dt \right| \leq \lambda \int_a^x |f(t)| \, dt = \lambda F(x).$$

Therefore

$$F'(x) = |f(x)| \leq \lambda F(x), \quad F(a) = 0, \quad F \geq 0.$$

Gronwall's inequality gives $F \equiv 0$, and hence

$$f \equiv 0.$$

No division by f is needed.

Example 2: IMC 1994, Problem 2

Let $a < b$ be real numbers and $f \in C^1((a, b))$. Suppose

$$\lim_{x \rightarrow a^+} f(x) = +\infty, \quad \lim_{x \rightarrow b^-} f(x) = -\infty,$$

and

$$f'(x) + f(x)^2 \geq -1.$$

Prove that

$$b - a \geq \pi,$$

and give an example attaining equality.

Notes: I1994-1-2. IMC 1994, Day 1, Problem 2.

Example 2: an arctangent substitution

Rewrite the inequality as

$$\frac{f'(x)}{1 + f(x)^2} \geq -1.$$

Example 2: an arctangent substitution

Rewrite the inequality as

$$\frac{f'(x)}{1 + f(x)^2} \geq -1.$$

Set

$$H(x) = x + \arctan f(x).$$

Then $H'(x) \geq 0$, so H is nondecreasing.

Example 2: an arctangent substitution

Rewrite the inequality as

$$\frac{f'(x)}{1 + f(x)^2} \geq -1.$$

Set

$$H(x) = x + \arctan f(x).$$

Then $H'(x) \geq 0$, so H is nondecreasing.

Taking the endpoint limits,

$$a + \frac{\pi}{2} \leq b - \frac{\pi}{2}.$$

Hence

$$b - a \geq \pi.$$

Example 2: an arctangent substitution

Rewrite the inequality as

$$\frac{f'(x)}{1 + f(x)^2} \geq -1.$$

Set

$$H(x) = x + \arctan f(x).$$

Then $H'(x) \geq 0$, so H is nondecreasing.

Taking the endpoint limits,

$$a + \frac{\pi}{2} \leq b - \frac{\pi}{2}.$$

Hence

$$b - a \geq \pi.$$

Equality: $f(x) = \cot(x - a)$ on $(a, a + \pi)$, since $f' + f^2 = -1$.

Concavity and tangent lines

Tangent-line inequality for concave functions

If u is differentiable and concave on an interval, then

$$u(y) \leq u(x) + u'(x)(y - x)$$

for all x, y in that interval.

Concavity and tangent lines

Tangent-line inequality for concave functions

If u is differentiable and concave on an interval, then

$$u(y) \leq u(x) + u'(x)(y - x)$$

for all x, y in that interval.

Consequence

Every positive, differentiable, concave function on $\mathbb{R}$ is constant.

Concavity and tangent lines

Tangent-line inequality for concave functions

If u is differentiable and concave on an interval, then

$$u(y) \leq u(x) + u'(x)(y - x)$$

for all x, y in that interval.

Consequence

Every positive, differentiable, concave function on $\mathbb{R}$ is constant.

If $u'(x) \neq 0$, the tangent line becomes negative in one direction, contradicting $u > 0$.

Example 3: IMC 2020, Problem 5

Find all positive C^2 functions $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfying

$$f(x)f''(x) \geq 2(f'(x))^2 \quad (x \in \mathbb{R}).$$

Example 3: IMC 2020, Problem 5

Find all positive C^2 functions $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfying

$$f(x)f''(x) \geq 2(f'(x))^2 \quad (x \in \mathbb{R}).$$

Which function has a second derivative involving

$$2(f')^2 - ff''?$$

Notes: I2020-2-5. IMC 2020, Day 2, Problem 5.

Example 3: take the reciprocal

Set $u = 1/f$. Since $f > 0$,

$$u > 0, \quad u'' = \frac{2(f')^2 - ff''}{f^3} \leq 0.$$

Example 3: take the reciprocal

Set $u = 1/f$. Since $f > 0$,

$$u > 0, \quad u'' = \frac{2(f')^2 - ff''}{f^3} \leq 0.$$

Thus u is positive and concave on $\mathbb{R}$. It must be constant.

Consequently, all solutions are

$$f(x) \equiv C, \quad C > 0.$$

Example 3: take the reciprocal

Set $u = 1/f$. Since $f > 0$,

$$u > 0, \quad u'' = \frac{2(f')^2 - ff''}{f^3} \leq 0.$$

Thus u is positive and concave on $\mathbb{R}$. It must be constant.

Consequently, all solutions are

$$f(x) \equiv C, \quad C > 0.$$

The domain matters: on $(0, \infty)$, the nonconstant function $f(x) = 1/x$ satisfies equality.

Taylor's theorem with integral remainder

Taylor's theorem

Let $f \in C^2(I)$, where I is an interval. If $x, x + h \in I$, then

$$f(x + h) = f(x) + hf'(x) + h^2 \int_0^1 (1 - t)f''(x + th) dt.$$

Taylor's theorem with integral remainder

Taylor's theorem

Let $f \in C^2(I)$, where I is an interval. If $x, x + h \in I$, then

$$f(x + h) = f(x) + hf'(x) + h^2 \int_0^1 (1 - t)f''(x + th) dt.$$

This follows by integration by parts.

Taylor's theorem with integral remainder

Taylor's theorem

Let $f \in C^2(I)$, where I is an interval. If $x, x + h \in I$, then

$$f(x + h) = f(x) + hf'(x) + h^2 \int_0^1 (1 - t)f''(x + th) dt.$$

This follows by integration by parts.

If $|f''| \leq L$ between x and $x + h$, then

$$|f(x + h) - f(x) - hf'(x)| \leq \frac{L}{2}h^2.$$

The formula holds for positive and negative h .

Taylor estimates with a Lipschitz derivative

Quadratic remainder estimate

Let $f : I \rightarrow \mathbb{R}$ be differentiable and suppose

$$|f'(x) - f'(y)| \leq L|x - y| \quad (x, y \in I).$$

Then, whenever $x, x + h \in I$,

$$|f(x + h) - f(x) - hf'(x)| \leq \frac{L}{2}h^2.$$

Taylor estimates with a Lipschitz derivative

Quadratic remainder estimate

Let $f : I \rightarrow \mathbb{R}$ be differentiable and suppose

$$|f'(x) - f'(y)| \leq L|x - y| \quad (x, y \in I).$$

Then, whenever $x, x + h \in I$,

$$|f(x + h) - f(x) - hf'(x)| \leq \frac{L}{2}h^2.$$

Proof.

$$f(x + h) - f(x) - hf'(x) = h \int_0^1 (f'(x + th) - f'(x)) \, dt.$$

Its absolute value is at most $Lh^2 \int_0^1 t \, dt = Lh^2/2$.

A second derivative is not required.

Exercise: IMC 2017, Problem 2

Let $f : \mathbb{R} \rightarrow (0, \infty)$ be differentiable. Suppose f' is L -Lipschitz, where $L > 0$:

$$|f'(x) - f'(y)| \leq L|x - y|.$$

Prove that

$$(f'(x))^2 < 2Lf(x) \quad \text{for every } x \in \mathbb{R}.$$

Exercise: IMC 2017, Problem 2

Let $f : \mathbb{R} \rightarrow (0, \infty)$ be differentiable. Suppose f' is L -Lipschitz, where $L > 0$:

$$|f'(x) - f'(y)| \leq L|x - y|.$$

Prove that

$$(f'(x))^2 < 2Lf(x) \quad \text{for every } x \in \mathbb{R}.$$

Hint: choose the step h in the Taylor estimate.

Notes: I2017-1-2. IMC 2017, Day 1, Problem 2.

IMC 2017: choose the step

The Taylor estimate gives

$$f(x+h) \leq f(x) + hf'(x) + \frac{L}{2}h^2.$$

IMC 2017: choose the step

The Taylor estimate gives

$$f(x+h) \leq f(x) + hf'(x) + \frac{L}{2}h^2.$$

Complete the square:

$$f(x) + hf'(x) + \frac{L}{2}h^2 = f(x) - \frac{(f'(x))^2}{2L} + \frac{L}{2} \left(h + \frac{f'(x)}{L} \right)^2.$$

IMC 2017: choose the step

The Taylor estimate gives

$$f(x+h) \leq f(x) + hf'(x) + \frac{L}{2}h^2.$$

Complete the square:

$$f(x) + hf'(x) + \frac{L}{2}h^2 = f(x) - \frac{(f'(x))^2}{2L} + \frac{L}{2} \left(h + \frac{f'(x)}{L} \right)^2.$$

Choose

$$h = -\frac{f'(x)}{L}.$$

The domain $\mathbb{R}$ allows this step for every x .

IMC 2017: the strict inequality

With $h = -f'(x)/L$,

$$0 < f\left(x - \frac{f'(x)}{L}\right) \leq f(x) - \frac{(f'(x))^2}{2L}.$$

Therefore

$$(f'(x))^2 < 2Lf(x).$$

IMC 2017: the strict inequality

With $h = -f'(x)/L$,

$$0 < f\left(x - \frac{f'(x)}{L}\right) \leq f(x) - \frac{(f'(x))^2}{2L}.$$

Therefore

$$(f'(x))^2 < 2Lf(x).$$

The strict inequality comes from $f > 0$.

If only $f \geq 0$ is assumed, we obtain

$$(f')^2 \leq 2Lf,$$

with equality for $f(x) = \frac{L}{2}x^2$.

Transformations used in this chapter

Expression	Transformation or identity
$f' - pf$	$(e^{-A}f)' = e^{-A}(f' - pf), \quad A' = p$
$f' + f^2 \geq -1$	$(x + \arctan f)' = \frac{1 + f^2 + f'}{1 + f^2} \geq 0$
$ff'' - 2(f')^2$	$(1/f)'' = -\frac{ff'' - 2(f')^2}{f^3}$
f'/f	$(\log f)' = f'/f$

Reciprocals and logarithms require $f \neq 0$.

The sign of the multiplier determines whether an inequality is preserved.

Further example: IMC 2023, Problem 1

Find all C^2 functions $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfying

$$f(7x + 1) = 49f(x) \quad (x \in \mathbb{R}).$$

Further example: IMC 2023, Problem 1

Find all C^2 functions $f : \mathbb{R} \rightarrow \mathbb{R}$ satisfying

$$f(7x + 1) = 49f(x) \quad (x \in \mathbb{R}).$$

The affine map $x \mapsto 7x + 1$ has the fixed point

$$x_* = -\frac{1}{6}.$$

Translate this point to the origin.

Notes: I2023-1-1. IMC 2023, Day 1, Problem 1.

IMC 2023: translate and differentiate

Set

$$g(t) = f\left(t - \frac{1}{6}\right).$$

Then

$$g(7t) = 49g(t).$$

IMC 2023: translate and differentiate

Set

$$g(t) = f\left(t - \frac{1}{6}\right).$$

Then

$$g(7t) = 49g(t).$$

Differentiating twice gives

$$g''(7t) = g''(t).$$

IMC 2023: translate and differentiate

Set

$$g(t) = f\left(t - \frac{1}{6}\right).$$

Then

$$g(7t) = 49g(t).$$

Differentiating twice gives

$$g''(7t) = g''(t).$$

For every fixed t ,

$$g''(t) = g''\left(\frac{t}{7^n}\right) \longrightarrow g''(0).$$

Thus g'' is constant.

IMC 2023: determine the solutions

Since g'' is constant,

$$g(t) = At^2 + Bt + C.$$

Substituting into $g(7t) = 49g(t)$,

$$49At^2 + 7Bt + C = 49At^2 + 49Bt + 49C.$$

IMC 2023: determine the solutions

Since g'' is constant,

$$g(t) = At^2 + Bt + C.$$

Substituting into $g(7t) = 49g(t)$,

$$49At^2 + 7Bt + C = 49At^2 + 49Bt + 49C.$$

Hence $B = C = 0$, and

$$f(x) = A \left(x + \frac{1}{6} \right)^2, \quad A \in \mathbb{R}.$$

Every such function satisfies the original equation.

A product substitution: Putnam 2009, A2

Near $x = 0$, differentiable functions f, g, h satisfy

$$\begin{aligned}f' &= 2f^2gh + \frac{1}{gh}, \\g' &= fg^2h + \frac{4}{fh}, \\h' &= 3fgh^2 + \frac{1}{fg},\end{aligned}\quad f(0) = g(0) = h(0) = 1.$$

Find $f(x)$ near 0.

A product substitution: Putnam 2009, A2

Near $x = 0$, differentiable functions f, g, h satisfy

$$\begin{aligned}f' &= 2f^2gh + \frac{1}{gh}, \\g' &= fg^2h + \frac{4}{fh}, \\h' &= 3fgh^2 + \frac{1}{fg},\end{aligned}\quad f(0) = g(0) = h(0) = 1.$$

Find $f(x)$ near 0.

Substitution: $p = fgh$.

Notes: P2009-A2. Putnam 2009, A2.

Putnam 2009: reduce to one equation

By the product rule,

$$p' = f'gh + fg'h + fgh' = (2p^2 + 1) + (p^2 + 4) + (3p^2 + 1).$$

Thus

$$p' = 6(p^2 + 1), \quad p(0) = 1.$$

Putnam 2009: reduce to one equation

By the product rule,

$$p' = f'gh + fg'h + fgh' = (2p^2 + 1) + (p^2 + 4) + (3p^2 + 1).$$

Thus

$$p' = 6(p^2 + 1), \quad p(0) = 1.$$

Consequently,

$$\arctan p(x) = 6x + \frac{\pi}{4}, \quad p(x) = \tan \left(6x + \frac{\pi}{4} \right)$$

on a sufficiently small neighbourhood of zero.

Putnam 2009: reduce to one equation

By the product rule,

$$p' = f'gh + fg'h + fgh' = (2p^2 + 1) + (p^2 + 4) + (3p^2 + 1).$$

Thus

$$p' = 6(p^2 + 1), \quad p(0) = 1.$$

Consequently,

$$\arctan p(x) = 6x + \frac{\pi}{4}, \quad p(x) = \tan \left(6x + \frac{\pi}{4} \right)$$

on a sufficiently small neighbourhood of zero.

Also,

$$\frac{f'}{f} = 2p + \frac{1}{p}.$$

Putnam 2009: recover f

Write $\theta = 6x + \pi/4$. Then

$$\frac{f'}{f} = 2 \tan \theta + \cot \theta.$$

Integrating and using $f(0) = 1$,

$$\log f(x) = -\frac{1}{3} \log \frac{\cos \theta}{\cos(\pi/4)} + \frac{1}{6} \log \frac{\sin \theta}{\sin(\pi/4)}.$$

Putnam 2009: recover f

Write $\theta = 6x + \pi/4$. Then

$$\frac{f'}{f} = 2 \tan \theta + \cot \theta.$$

Integrating and using $f(0) = 1$,

$$\log f(x) = -\frac{1}{3} \log \frac{\cos \theta}{\cos(\pi/4)} + \frac{1}{6} \log \frac{\sin \theta}{\sin(\pi/4)}.$$

Hence

$$f(x) = \left(\frac{\cos(\pi/4)}{\cos(6x + \pi/4)} \right)^{1/3} \left(\frac{\sin(6x + \pi/4)}{\sin(\pi/4)} \right)^{1/6}.$$

All factors are positive on a sufficiently small neighbourhood of zero.